Week 5.

1. Classes

A class is a blueprint that holds variable names and functions specific to that class. A class itself doesn't store the variable and functions in memory, it is just a blueprint.

When you use that blueprint to build a class instance, you make an 'object'.

Eg:

```
class Student:
    Name
    DOB
    Program
    Mark ↑↓
    grade()
    getAge()
```

create a class instance:

```
std = Student()
print(std)
```

↳ prints memory address where class instance std is stored.

```
std.name = "John"
std.mark = 92
print(std.name) # John
print(std.mark) # 92
```

2. Constructors

classes require constructor function '__init__' to initialize variables. E.g:

```
class Student:
    def __init__(self, name, dob):
        self.name = name
        self.dob = dob

    def getName(self):
        print(self.name)
```